

A Unified Framework for Adsorption Isotherm Analysis in Water Quality Studies: Descriptors, Model Selection, and Reporting Guidelines

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Abstract

Adsorption isotherms are central to water-quality research, yet their fitted parameters are often compared across models as if they were physically interchangeable. This practice can be misleading because common isotherm equations differ in their asymptotic behavior, parameter dimensions, and assumptions about surface heterogeneity. Here, we propose a unified framework for adsorption isotherm analysis aimed at making capacity, affinity, and heterogeneity descriptors comparable across adsorbents, contaminants, and studies. The framework separates fitted model parameters from reportable physical descriptors and defines a minimum descriptor set consisting of the saturation capacity $Q_{\max}$, a dimensionless standard-state affinity K_{aff}^* , the corresponding adsorption-energy descriptor E_{ads} , and a reported heterogeneity width σ_H . We show that $Q_{\max}$ should be reported only for models with a finite saturation plateau, whereas non-saturating models such as Freundlich cannot provide a physically meaningful maximum capacity. Model-dependent affinity constants are recast through a declared standard-state concentration C^0 , allowing thermodynamic comparison on a common scale. Surface heterogeneity is treated through the energy distributions implied by each isotherm family. In addition, we provide a practical workflow combining non-linear regression, residual metrics, information criteria, F-tests, physical decision rules, and optional KNN/SVM-based material classification. Together, these guidelines provide a reproducible route from raw isotherm data to statistically validated, physically interpretable, and cross-study comparable adsorption descriptors for water-treatment materials.

1 Introduction

Access to safe water is becoming one of the defining challenges of this century. Global population reached approximately 8.2 billion in 2024 and is projected to approach 9.7 billion by 2050.¹ Yet, 2.2 billion people still lack safely managed drinking water services.² In addition, roughly half of the world’s population experiences severe water scarcity for at least part of

the year.³ Urbanization, industrial activity, agricultural intensification, and climate change are compounding these pressures on global water systems.⁴ Within this context, the United Nations Sustainable Development Goal 6 (SDG 6), “Clean Water and Sanitation,” has catalyzed broad scientific and policy interest in water treatment technologies. However, recent global assessments indicate that the world is not on track to achieve SDG 6 by 2030.⁵

Among the available remediation strategies, adsorption remains especially attractive because of its operational simplicity, compatibility with decentralized treatment, and applicability across a wide range of contaminant classes. Historically, adsorption research in water purification has focused on metals, metalloids, dyes, and pesticides.^{6–8} The contaminant landscape, however, is expanding rapidly and now extends far beyond these classical targets. Contemporary challenges increasingly involve persistent and mobile organic pollutants, pharmaceutical residues, endocrine-disrupting compounds, microplastic-associated chemicals, and, most prominently, per- and polyfluoroalkyl substances (PFAS).^{9–12} PFAS have become emblematic of the next generation of adsorption targets: they are highly persistent, often mobile in water, resistant to conventional degradation, and subject to tightening regulatory limits in drinking water worldwide.^{13–15} As the diversity of target contaminants grows, so does the need for rigorous, cross-pollutant comparison of adsorbent performance, a task that demands a common descriptive language for adsorption data.^{16,17}

Adsorption isotherms are the standard tool for summarizing material performance and for extracting descriptors such as maximum uptake capacity, affinity, and site heterogeneity. In practice, however, parameters obtained from different isotherm models are routinely tabulated side by side as though they were interchangeable. They are not. The Langmuir, double Langmuir, Freundlich, and Sips equations arise from different physical assumptions and assign different meanings, dimensions, and asymptotic behaviors to their fitted constants.^{18–20} Comparing a Langmuir $Q_{\max}$ with a value extrapolated from a Freundlich fit, or placing a Langmuir affinity constant next to a Sips affinity constant whose units depend on the heterogeneity exponent, is neither mathematically consistent nor physically meaning-

55 ful.^{21,22} What often appears to be a comparison of materials is, in reality, a comparison of
56 incompatible parameterizations.¹⁶

57 This problem is not merely academic. As adsorption studies move toward application-
58 driven screening of advanced sorbents for complex water matrices—including PFAS, phar-
59 maceuticals, and other emerging contaminants—materials are increasingly ranked as “better”
60 on the basis of larger fitted constants derived from different models, concentration scales,
61 and assumptions about site heterogeneity. Such practice weakens reproducibility, complicates
62 translation from laboratory results to engineering design, and, most importantly, prevents the
63 community from building reliable structure–performance relationships across adsorbents and
64 pollutants. A field that aims to address global water challenges, including those highlighted
65 by SDG 6 and the rapidly evolving contaminant landscape, requires reporting standards that
66 make adsorption parameters comparable in a thermodynamically meaningful way.

67 In this work, we propose a practical reporting framework grounded in four principles.
68 First, capacity comparisons should be restricted to isotherm models that define a finite
69 saturation capacity. Second, model-dependent affinity constants should be converted, when-
70 ever possible, into a dimensionless standard-state affinity and the corresponding adsorption-
71 energy descriptor. Third, model selection should be based on residual diagnostics, informa-
72 tion criteria, nested-model tests, and physical admissibility, rather than on R^2 alone. Fourth,
73 once reliable descriptors have been extracted, they can be used for optional downstream clas-
74 sification or clustering of adsorption systems. The mathematical derivations are provided in
75 the SI, and the companion notebooks illustrate the workflow with synthetic examples and
76 user-supplied data.

2 The problem: incompatibility of isotherm-derived parameters

2.1 Common isotherm models and their assumptions

The most widely used isotherm equations in aqueous-phase adsorption can be written as follows. A complete list is included in Table 1 and SI.

Langmuir. Assumes a homogeneous surface with a single class of equivalent sites and monolayer coverage:

$$q_e = \frac{Q_{\max} K_L C_e}{1 + K_L C_e}. \quad (1)$$

Parameters: $Q_{\max}$ (mg g^{-1}), the finite saturation capacity, and K_L (L mg^{-1}), the affinity constant. See Eq. (S-3).

Double Langmuir. Extends Langmuir to two independent site populations:

$$q_e = \frac{Q_1 K_1 C_e}{1 + K_1 C_e} + \frac{Q_2 K_2 C_e}{1 + K_2 C_e}. \quad (2)$$

Total capacity $Q_{\max} = Q_1 + Q_2$ remains finite and well defined. See Eq. (S-14).

Freundlich. An empirical power law that describes heterogeneous adsorption without a saturation plateau:

$$q_e = K_F C_e^{1/n}. \quad (3)$$

K_F has concentration-dependent units $(\text{mg g}^{-1})(\text{L mg}^{-1})^{1/n}$; no $Q_{\max}$ is defined. See Eq. (S-22).

91 **Sips (Langmuir–Freundlich).** Introduces surface heterogeneity through an exponent m
 92 while retaining a saturation plateau:

$$q_e = \frac{Q_{\max} K_S C_e^m}{1 + K_S C_e^m}. \quad (4)$$

93 When $m = 1$, Sips reduces to Langmuir. The affinity constant K_S carries units that depend
 94 on m , i.e. $(\text{L mg}^{-1})^m$, and is therefore not directly comparable with K_L . See Eq. (S-29).

Parameterization note. In this manuscript, the Sips constant is defined outside the concentration exponent, $K_S C_e^m$, consistently in the main text, SI, and fitting code. With this convention K_S carries units of $[C]^{-m}$, and the reciprocal-concentration affinity used for descriptor extraction is $K_S^{1/m}$.

95 As shown in Figures 1–3, the model selection procedure combines physical screening,
 96 statistical comparison, and physicochemical validation.

Table 1: Availability of $Q_{\max}$, K_H , and an effective affinity constant K_{aff} for common isotherm models, together with their implied energy-distribution family. The “Recommended use” column is **YES** when both $Q_{\max}$ and a comparable K_{aff} can be reported, and **NO** otherwise. See the SI sections in the last column for the derivations and definitions.

Model	$Q_{\max}$	K_H	K_{aff}	Recommended	Energy distribution	SI ref.
Henry	No	Yes	No	NO	– (low-coverage limit)	S-1.1
Langmuir	Yes	Yes	Yes	YES	Dirac δ	S-1.2
Double Langmuir	Yes	Yes	Yes	YES	Two Dirac δ	S-1.3
Freundlich	No	Cond.	No	NO	Power law	S-1.4
Sips (L–F)	Yes	Cond.	Yes *	YES	Logistic	S-1.5
Tóth	Yes	Yes	Yes	YES	Gen. logistic (asym.)	S-1.6
Jovanovic	Yes	Yes	Yes	YES	Gumbel (EV-I)	S-1.7
Temkin	No	No	Yes *	NO	Uniform (rect.)	S-1.8
UNILAN	Yes	Yes *	Yes	YES	Uniform (rect.)	S-1.9
Redlich– Peterson	Cond.	Yes	Yes *	NO	Non-standard (asym.)	S-1.10
Koble– Corrigan	Yes	Cond.	Yes *	YES	Logistic (Sips like)	S-1.11
Radke– Prausnitz	No	Yes	Cond. *	NO	Non-standard (asym.)	S-1.12
D-R / D-A	Yes	No	Yes *	YES	Rayleigh / Weibull	S-1.13
Hill	Yes	Cond.	Yes *	YES	Logistic (Sips like)	S-1.14

⁹⁷ **Notes.** “Cond.” indicates availability only in special parameter limits (e.g., $m = 1$, $n_H = 1$,
⁹⁸ $\beta = 1$). “*.” indicates an effective or operational definition; see the SI for explicit formulas
⁹⁹ and limits.

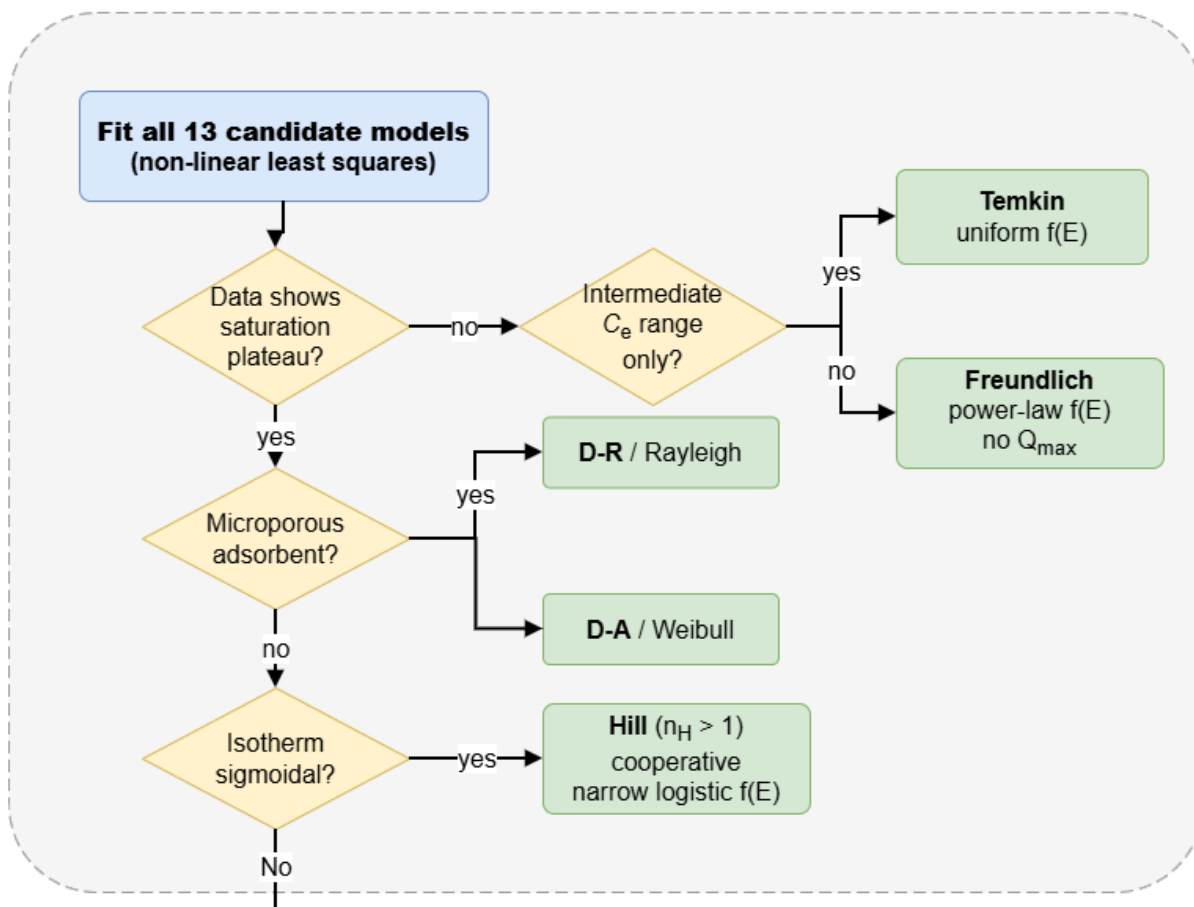


Figure 1: Shape-based pre-screening of adsorption isotherm models. Experimental isotherms are first fitted with the diagnostic model library by non-linear least squares and then classified according to physically observable features, including the presence of a saturation plateau, microporous uptake behavior, sigmoidal adsorption, and restriction to an intermediate concentration range. This first decision layer separates models that can provide finite capacity descriptors from models that are better interpreted as local, empirical, or special-regime descriptions.

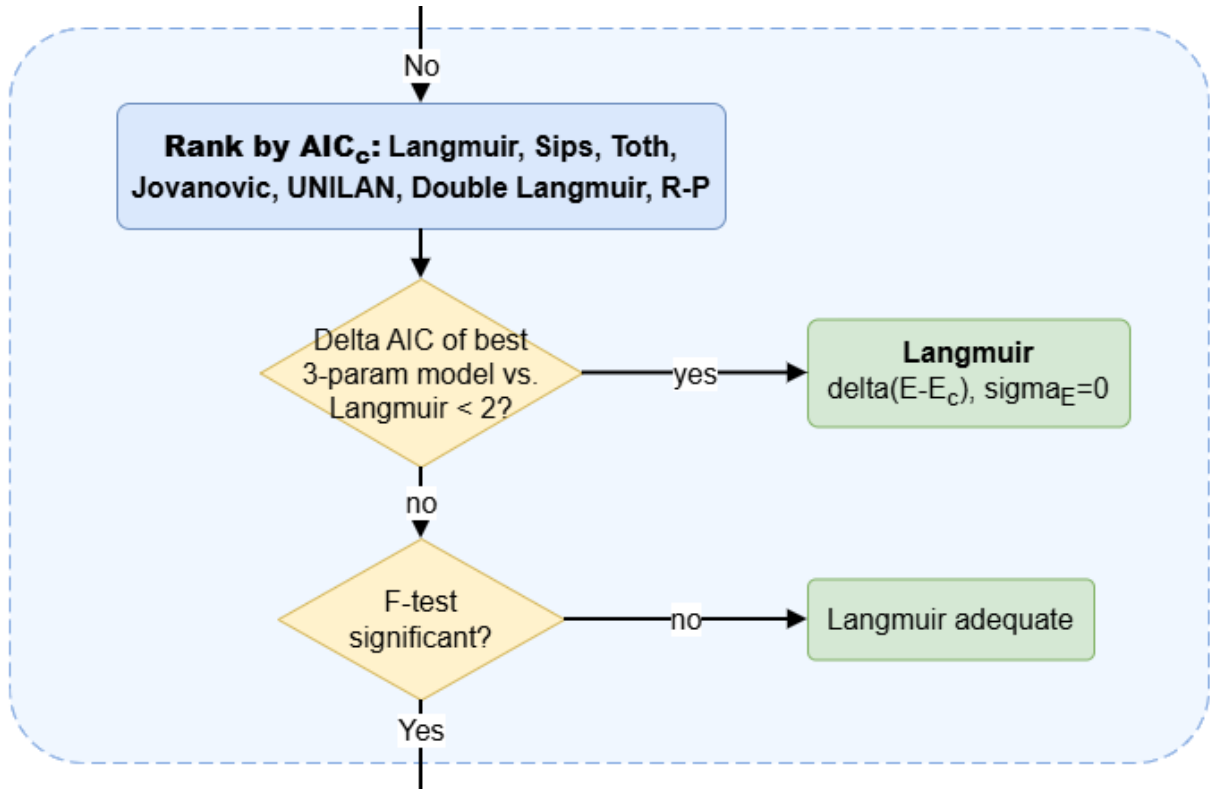


Figure 2: Statistical model-selection step used to decide whether a model more complex than Langmuir is justified. Candidate saturating models are ranked by the corrected Akaike information criterion (AIC_c), and the best three-parameter model is compared against Langmuir. When the improvement in AIC_c is small or the F-test is not significant, the Langmuir model is retained as an adequate description, avoiding over-interpretation of weak heterogeneity.

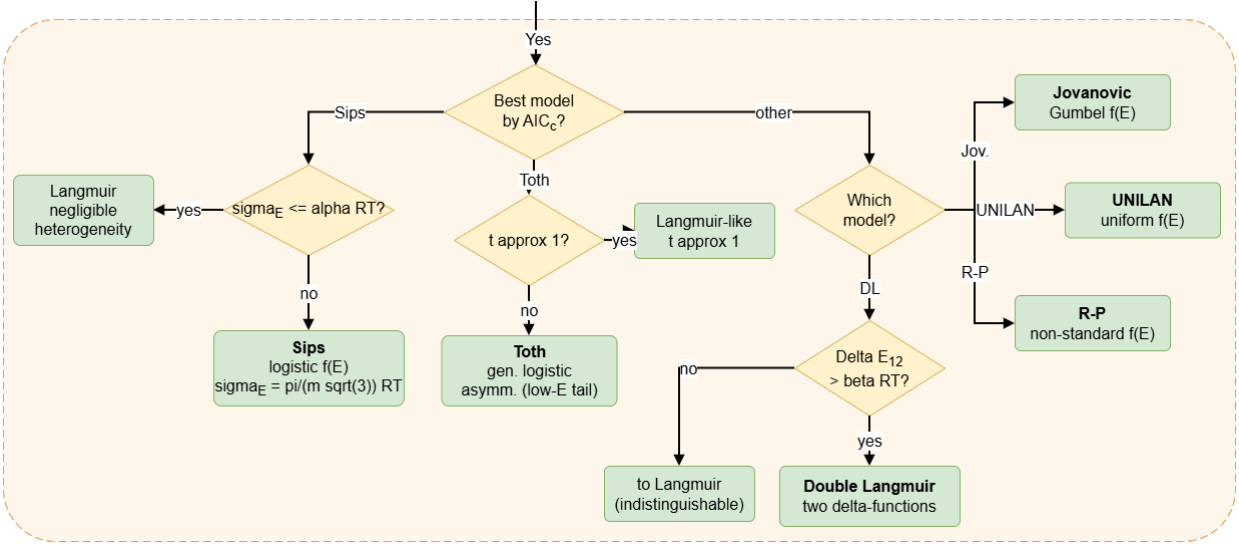


Figure 3: Final physical-validation step for assigning the selected isotherm model to a reportable adsorption descriptor set. The best AIC_c model is interpreted in terms of its implied energy-distribution family and limiting behavior. Sips and Tóth fits are checked for proximity to the Langmuir limit, while Jovanovic, UNILAN, Redlich–Peterson, and double-Langmuir outcomes are assigned according to their characteristic heterogeneity signatures. This step links statistical goodness of fit with physically meaningful descriptors such as finite capacity, affinity, and energy-distribution width.

2.2 Why $Q_{\max}$ values are not always comparable

Even though $Q_{\max}$ is often treated as a universal figure of merit, it is model-specific unless the isotherm explicitly contains a finite saturation plateau. In the Langmuir and Double Langmuir equations, $Q_{\max}$ (or $Q_1 + Q_2$) is a true asymptote: it is the coverage reached as $C_e \rightarrow \infty$ and is therefore a well-defined thermodynamic capacity. By contrast, the Freundlich isotherm has no saturation term; q_e increases without bound as C_e increases (SI Eq. (S-22)), so any “ $Q_{\max}$ ” inferred from a Freundlich fit depends entirely on the chosen concentration window. Reporting such a value as a capacity is physically meaningless because it is not a model parameter and has no invariant interpretation.

Even for models that do saturate, comparability depends on whether the saturation regime is actually constrained by the experimental data. In the Sips equation, for example, the heterogeneity exponent m can shift the onset of saturation far beyond the measured

concentration range (see Eq. S-29). Under these conditions, the fitted $Q_{\max}$ is not determined by the data but inferred through extrapolation. As a result, reported capacities may reflect model-imposed asymptotes rather than experimentally supported saturation values. The practical consequence is that tables mixing Langmuir or Double Langmuir $Q_{\max}$ values with Freundlich-derived or poorly constrained Sips capacities are not comparing the same physical quantity. Any ranking built on such a mixture conflates experimentally supported saturation capacities with model-dependent extrapolations.

2.3 Why affinity constants are not comparable across models

Affinity constants carry the strongest illusion of comparability because they are reported as single numbers, yet their physical meaning and units change from model to model. In the Langmuir equation, K_L has units of inverse concentration (e.g., L mg^{-1}) and sets the midpoint of the uptake curve. In the Sips equation, the constant K_S carries units of $(\text{L mg}^{-1})^m$, which depend on the heterogeneity exponent m ; therefore its numerical value is not on the same scale as K_L unless $m = 1$ (SI Eqs. (S-3) and (S-29)). The Freundlich constant K_F is even less comparable, because its units depend on $1/n$ and it does not correspond to any saturation behavior (SI Eq. (S-22)).

The Henry constant K_H provides a low-coverage slope (SI Eq. (S-7)), but it is not a full-isotherm descriptor and cannot replace affinity constants derived from different functional forms without additional normalization. As a result, placing K_L , K_S , and K_F side by side implies a common meaning that does not exist. Any comparison based on raw affinity constants is therefore a comparison of incommensurate parameters, not of intrinsic adsorption strength.

3 A common language: standard-state adsorption energy

3.1 Constructing a dimensionless equilibrium constant

To place all isotherm models on a common thermodynamic footing, the model-specific affinity parameter must be expressed as a dimensionless equilibrium constant. This is necessary because the fitted affinity constants used in adsorption isotherms are usually dimensional quantities whose units depend on the mathematical form of the model and on the units chosen for concentration and uptake. Therefore, direct comparison of fitted affinity parameters between different models can be misleading unless a common normalization procedure is applied.

Here, the affinity is defined from the low-concentration Henry region of the isotherm. The Henry constant is obtained as the initial slope,

$$K_H = \lim_{C_e \rightarrow 0} \left(\frac{q_e}{C_e} \right), \quad (5)$$

where q_e is the equilibrium uptake and C_e is the equilibrium concentration in solution. For example, if q_e is expressed in mg g^{-1} and C_e in mg L^{-1} , then

$$[K_H] = \frac{\text{mg g}^{-1}}{\text{mg L}^{-1}} = \text{L g}^{-1}. \quad (6)$$

Since K_H still contains the adsorbent mass dependence, it is normalized by the maximum adsorption capacity $Q_{\max}$. If $Q_{\max}$ is expressed in mg g^{-1} , then

$$\left[\frac{K_H}{Q_{\max}} \right] = \frac{\text{L g}^{-1}}{\text{mg g}^{-1}} = \text{L mg}^{-1}. \quad (7)$$

Thus, $K_H/Q_{\max}$ has units of inverse concentration. A dimensionless affinity constant is then obtained by multiplying this quantity by a declared standard-state concentration C^0

expressed in the same concentration units as C_e :

$$K_{\text{aff}}^* = \frac{K_H}{Q_{\text{max}}} C^0. \quad (8)$$

For instance, if C^0 is expressed in mg L^{-1} , then

$$[K_{\text{aff}}^*] = (\text{L mg}^{-1}) (\text{mg L}^{-1}) = 1. \quad (9)$$

Therefore, K_{aff}^* is dimensionless by construction. For comparisons among different adsorbates, the declared standard state should preferably be expressed on a molar concentration scale or, when available, in terms of activity. Concentrations in mg L^{-1} are acceptable for water-quality reporting and engineering screening, but they should be interpreted as operational descriptors rather than as composition-independent thermodynamic standard states. For models such as Sips or Koble–Corrigan, where the initial slope at $C_e \rightarrow 0$ diverges (i.e., a finite Henry constant does not exist), an operational affinity constant is defined instead by raising the native affinity parameter (carrying fractional units) to the power of the reciprocal exponent (e.g., $K_S^{1/m}$), thereby restoring the units of inverse concentration before normalization.

In this construction, Q_{max} removes the dependence of the Henry slope on the total adsorption capacity, while C^0 removes the remaining concentration unit. Thus, K_{aff}^* represents the standard-state-normalized affinity of the adsorbent–adsorbate pair relative to a defined concentration standard. For the Langmuir isotherm, for example, $K_H = Q_{\text{max}} K_L$, and therefore $K_{\text{aff}}^* = K_L C^0$, recovering the usual dimensionless form of the Langmuir equilibrium constant. The same procedure can be applied to any isotherm model for which a finite Henry limit and a meaningful saturation capacity can be defined.

It should be emphasized that the numerical value of K_{aff}^* depends on the chosen standard state C^0 . Therefore, C^0 must be explicitly reported. However, once the same C^0 is used consistently, the resulting dimensionless constants allow affinity parameters extracted

173 from different isotherm models to be compared on the same basis and used in standard
 174 thermodynamic expressions.

175 **3.2 Standard-state adsorption energy E_{ads}**

176 The standard Gibbs free energy associated with the declared standard state is

$$\Delta G_{\text{ads}}^0 = -RT \ln K_{\text{aff}}^*. \quad (10)$$

177 For reporting, we use the positive affinity-energy descriptor

$$E_{\text{ads}} \equiv E_{\text{aff}} = -\Delta G_{\text{ads}}^0 = RT \ln K_{\text{aff}}^*. \quad (11)$$

178 Thus, larger E_{ads} corresponds to stronger standard-state affinity. Because K_{aff}^* is dimension-
 179 less, the logarithmic term is dimensionless and the units of E_{ads} arise only from RT :

$$[E_{\text{ads}}] = [RT] = \text{kJ mol}^{-1} \quad (12)$$

180 when R is expressed in $\text{kJ mol}^{-1} \text{K}^{-1}$. The descriptor is therefore a standardized model-
 181 derived measure of affinity, not a raw fitted parameter and not a directly measured observable
 182 independent of the fitted model.

183 **3.3 Extraction of K_{aff}^* and E_{ads} from each model**

184 Table 2 shows how K_{aff}^* and E_{ads} are obtained from the native parameters of each isotherm.

Table 2: Extraction of native and operational affinity descriptors from isotherm parameters. C^0 is the declared standard-state concentration; descriptors are reportable only when the fitted model is physically admissible and identifiable.

Model	$Q_{\max}$	K_{aff} (native)	$E_{\text{ads}}/(RT)$
Langmuir	$Q_{\max}$	K_L	$\ln(K_L C^0)$
Double Langmuir	$Q_1 + Q_2$	$K_L^{(\text{tot})} = \frac{Q_1 K_1 + Q_2 K_2}{Q_1 + Q_2}$	$\ln(K_L^{(\text{tot})} C^0)$
Sips	$Q_{\max}$	$K_S^{1/m}$	$\frac{1}{m} \ln(K_S (C^0)^m)$
Tóth	$q_{\max}$	b	$\ln(b C^0)$
Jovanovic	$q_{\max}$	b	$\ln(b C^0)$
Temkin	—	a_T	$\ln(a_T C^0)$
UNILAN	$Q_{\max}$	$\sqrt{b_{\max} b_{\min}}$	$\frac{1}{2} \ln(b_{\max} b_{\min} (C^0)^2)$
Redlich–Peterson	K_{RP}/a_{RP} ^a	a_{RP} ^a	$\ln(a_{RP} C^0)$
Koble–Corrigan	A_{KC}/B_{KC}	B_{KC}^n	$n \ln(B_{KC} C^0)$
Radke–Prausnitz	—	r_{RP}	$\ln(r_{RP} C^0)$
D-R / D-A	$Q_{\max}$	$1/C_{1/2}$ ^b	$\ln(C^0/C_{1/2})$
Hill	$Q_{\max}$	K_D^{-1/n_H}	$-\frac{1}{n_H} \ln K_D + \ln C^0$

^a Only when $\beta = 1$. ^b $C_{1/2}$ is the half-loading concentration; see Sec. S-1.13. “—” indicates no finite $Q_{\max}$ exists.

185 The Freundlich isotherm may describe adsorption data over a restricted concentration
186 window, but it is absent from Table 2 because it defines no finite $Q_{\max}$. Consequently, the
187 ratio $K_H/Q_{\max}$ is undefined and no universal dimensionless affinity can be constructed from
188 the Freundlich parameters alone. Only a local, reference-dependent apparent affinity can be
189 defined after choosing a concentration C_{ref} , and this quantity should not be reported as a
190 universal descriptor. For the Double Langmuir model, an effective total adsorption energy
191 can be reported from the capacity-weighted constant $K_L^{(\text{tot})}$, while the site-resolved constants
192 K_1 and K_2 remain necessary to preserve the bimodal interpretation captured by the site-
193 resolved energy separation. Finally, the choice of standard-state concentration C^0 must be
194 explicitly declared; although it appears inside the logarithm and therefore cancels in any
195 difference ΔE_{ads} , it must be consistent across all entries in a comparative table. The full
196 set of energy-distribution families and their relation to σ_E is given in Table S-1; numerical
197 values of σ_E at $T = 298$ K are collected in Table S-2.

4 Quantifying surface heterogeneity: energy distributions

4.1 From adsorption sites to observable energy distributions

The final variable in the adsorption system is the distribution of site energies. Unlike $Q_{\max}$ or E_{ads} , this quantity is usually not fitted explicitly, yet it is implicitly encoded in every isotherm model. In other words, each adsorption equation corresponds to a particular assumption about how adsorption energies are distributed across the surface. This provides a natural way to place different isotherm models in a common physical framework.

Real adsorbent surfaces are energetically heterogeneous because adsorption sites differ in local chemistry, accessible geometry, coordination environment, and confinement. As a result, adsorption is more realistically described by a distribution of site energies than by a single characteristic value. A heterogeneous surface can therefore be represented as an ensemble of independent Langmuir-type sites, each characterized by an adsorption energy E , so that the macroscopic isotherm is written as

$$q_e = Q_{\max} \int_{-\infty}^{\infty} \Theta_L(E - E_c) f(E) dE, \quad (13)$$

where $f(E)$ is the normalized distribution of adsorption site energies,

$$\int_{-\infty}^{\infty} f(E) dE = 1, \quad (14)$$

and $\Theta_L(E - E_c)$ is the Langmuir occupation kernel,

$$\Theta_L(E - E_c) = \frac{1}{1 + \exp[-(E - E_c)/(RT)]}. \quad (15)$$

213 Here E_c is the condensation energy associated with the equilibrium concentration,

$$E_c = RT \ln \left(\frac{1}{K_0 C_e} \right), \quad (16)$$

214 with K_0 a reference constant used to define the energy scale.

215 Equation (13) emphasizes an important point: the experimentally observed isotherm
 216 does not probe the intrinsic distribution $f(E)$ directly, but rather the convolution of that
 217 distribution with the thermal occupation kernel. Differentiating eq (13) with respect to E_c
 218 gives the observable energy-density profile

$$\phi(E_c) = -\frac{1}{Q_{\max}} \frac{dq_e}{dE_c} = \int_{-\infty}^{\infty} f(E) g_T(E - E_c) dE, \quad (17)$$

219 where

$$g_T(x) = \frac{\exp(-x/RT)}{RT [1 + \exp(-x/RT)]^2} \quad (18)$$

220 is the thermal broadening kernel. This kernel is a logistic probability density with standard
 221 deviation

$$\sigma_T = \frac{\pi}{\sqrt{3}} RT. \quad (19)$$

222 Thus, even a perfectly homogeneous Langmuir surface, intrinsically described by a Dirac
 223 distribution,

$$f(E) = \delta(E - E_1), \quad (20)$$

224 is observed through the broadened profile

$$\phi(E_c) = g_T(E_1 - E_c), \quad (21)$$

225 which already has a finite width set by temperature. At $T = 298$ K, this intrinsic thermal
 226 width is

$$\sigma_T \approx 4.5 \text{ kJ mol}^{-1}. \quad (22)$$

227 This provides a physical interpretation for why systems with very small energetic dispersion
 228 often appear experimentally indistinguishable from Langmuir behavior.

229 The commonly used condensation approximation replaces the smooth Langmuir kernel
 230 by a step function,

$$\Theta_L(E - E_c) \approx \Theta(E - E_c), \quad (23)$$

231 so that thermal broadening is neglected and the intrinsic distribution can be recovered di-
 232 rectly as

$$f(E) = -\frac{1}{Q_{\max}} \frac{dq}{dE}. \quad (24)$$

233 This approximation is useful because it provides a practical route from any empirical isotherm
 234 equation back to its implied energy distribution. Full derivations for the models treated here
 235 are given in Section S-4.4.

236 **4.2 Discrete sites, thermal broadening, and continuous heterogene-** 237 **ity**

238 The relation between intrinsic and observable distributions clarifies how the usual isotherm
 239 families are connected. Consider first a surface with two discrete classes of sites,

$$f(E) = w \delta(E - E_1) + (1 - w) \delta(E - E_2), \quad 0 \leq w \leq 1. \quad (25)$$

240 Under thermal broadening, the experimentally accessible profile becomes

$$\phi(E_c) = w g_T(E_1 - E_c) + (1 - w) g_T(E_2 - E_c). \quad (26)$$

241 If the two site energies are sufficiently separated relative to the thermal broadening scale,
 242 the bimodal character is preserved and a Double Langmuir description remains physically
 243 meaningful. For the symmetric case of equal populations, the onset of bimodality for the

244 logistic kernel occurs at

$$|E_1 - E_2| > 2RT \operatorname{arcosh}(2) \approx 2.63 RT \approx 1.45 \sigma_T. \quad (27)$$

245 Below this scale, the two thermally broadened contributions overlap strongly and may appear
246 as a single broadened population. In that regime, the same data may be described by a
247 continuous heterogeneous model such as Sips or Tóth rather than by two explicitly resolved
248 Langmuir populations. For unequal populations, the exact threshold also depends on the
249 statistical weights of the two site classes.

250 This interpretation also explains the role of temperature. Increasing T increases the
251 kernel width σ_T according to eq (19), making nearby site populations less distinguishable.
252 Conversely, lower temperatures sharpen the energetic resolution and make multimodal struc-
253 ture easier to detect. Therefore, the observed heterogeneity is not only a property of the
254 surface itself, but also of the thermal scale at which the experiment is performed.

255 In this sense, the standard isotherm families can be viewed as limiting cases of the same
256 general picture. Langmuir corresponds to a single discrete energy level. Double Langmuir
257 corresponds to two discrete levels. Sips, Hill, and Koble–Corrigan correspond to symmetric
258 continuous distributions with finite width. Tóth and Jovanovic correspond to asymmet-
259 ric continuous distributions. Freundlich represents the limiting case of extremely broad or
260 effectively unbounded heterogeneity. The different models are therefore not disconnected em-
261 pirical formulas, but alternative representations of how adsorption energies are distributed
262 and thermally resolved.

263 **4.3 Energy-distribution families implied by common isotherms**

264 Applying eq (24) to each isotherm yields the energy-distribution families summarized in
265 Table 3. These distributions should be interpreted as the model-implied site-energy distri-
266 butions under the condensation approximation.

Table 3: Energy-distribution families and apparent mathematical width σ_E implied by each isotherm model. Parameter m (Sips, Koble–Corrigan) is the heterogeneity exponent; t (Tóth) and n_H (Hill) are the respective shape parameters. σ_E is in units of RT ($RT \approx 2.48 \text{ kJ mol}^{-1}$ at 298 K). Full derivations are given in Section S-4.4.

Model	Distribution $f(E)$	Symm.	$\sigma_E/(RT)$
Langmuir	Dirac $\delta(E - E_c)$	—	0
Double Langmuir	Two Dirac δ	—	$\frac{\sqrt{Q_1 Q_2}}{Q_1 + Q_2} \frac{ \Delta E_{12} }{RT}$
Sips	Logistic (scale RT/m)	Yes	$\pi/(m\sqrt{3})$
Tóth	Generalized logistic type IV	No	$> \pi/\sqrt{3}$ ($t < 1$); no closed form
Jovanovic	Gumbel (EV-I)	No	$\pi/\sqrt{6} \approx 1.28$
Temkin	Uniform	Yes	$b_T Q_{\max}/(2\sqrt{3} RT)$
UNILAN	Uniform (exact kernel)	Yes	$\ln(b_{\max}/b_{\min})/(2\sqrt{3})$
Freundlich	Power law (unbounded)	—	∞
Redlich–Peterson	Non-standard ($\beta < 1$)	No	—
Koble–Corrigan	$\equiv$ Sips	Yes	$\pi/(m\sqrt{3})$
Radke–Prausnitz	Non-standard	No	—
D-R ($n_{DA} = 2$)	Rayleigh	No	$0.463 E_a/(RT)$
D-A (general)	Weibull (shape n_{DA})	No	$(E_a/RT)\sqrt{\Gamma(1 + 2/n_{DA}) - [\Gamma(1 + 1/n_{DA})]^2}$
Hill	$\equiv$ Sips (scale RT/n_H)	Yes	$\pi/(n_H\sqrt{3})$

Several physically meaningful relations follow from Table 3. First, Langmuir and Double Langmuir represent discrete energetic populations, whereas Sips, Hill, Koble–Corrigan, Tóth, Jovanovic, Temkin, and UNILAN represent continuous distributions. Second, Sips, Hill, and Koble–Corrigan belong to the same symmetric logistic family and differ only in parameterization. Third, Tóth and Jovanovic introduce skewness, showing that heterogeneity is not fully described by width alone. Fourth, Freundlich corresponds to a limiting case in which the energy dispersion becomes effectively unbounded, consistent with the absence of a finite saturation capacity. Finally, the Dubinin family is naturally expressed in the Polanyi adsorption potential rather than directly in E , but still admits an equivalent probabilistic interpretation through Weibull-type distributions.

4.4 Physical and statistical interpretation of the energy distribution

Once an isotherm has been mapped onto an energy distribution, two widths must be kept conceptually separate. The quantity σ_E denotes the apparent mathematical width of the model-implied energy distribution. The reported heterogeneity descriptor is σ_H , which removes the homogeneous thermal contribution whenever that correction is defined. Therefore, Langmuir is recovered at $\sigma_H = 0$, even though thermally broadened occupation kernels may have a finite apparent width. Representative numerical values and the model-specific corrections are given in Table S-2.

This representation also provides a natural connection to statistical model comparison. Two different isotherm models may fit the same data well because their implied energy distributions have similar central energies and similar widths, even if their algebraic forms differ. For example, a narrow bimodal distribution and a single broader logistic distribution may generate nearly indistinguishable isotherms when the two discrete energies are closer than the thermal resolution. Conversely, two models with similar residual error but markedly different E_{ads} or σ_E imply different physical pictures of the surface.

The similarity between two implied energy distributions, $f_1(E)$ and $f_2(E)$, can be quantified directly through an overlap coefficient,

$$\text{OVL} = \int_{-\infty}^{\infty} \min[f_1(E), f_2(E)] dE, \quad 0 \leq \text{OVL} \leq 1. \quad (28)$$

High overlap indicates that two models encode essentially the same energetic picture, whereas low overlap indicates physically distinct heterogeneity patterns.

More broadly, the descriptor set

$$\mathcal{F} = \{Q_{\text{max}}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}. \quad (29)$$

defines a compact, physically interpretable feature space for adsorption systems. Within

298 this space, $Q_{\max}$ measures the amount of adsorption, E_{ads} measures the central energetic
 299 level, and σ_H measures the reported model-implied heterogeneity. These descriptors can
 300 be used not only to compare fitted models, but also to compare materials, contaminants,
 301 or operating conditions on a common basis. They also provide suitable input variables for
 302 data-driven analysis, including clustering and supervised classification methods such as k-
 303 nearest neighbours (KNN) and support vector machines (SVM), as developed in the following
 304 section. In this way, the energy-distribution view links isotherm physics, statistical model
 305 discrimination, and descriptor-based classification within a single framework.

306 5 Model selection and statistical comparison

307 5.1 Non-linear regression

308 To extract the proposed standardized reporting descriptors, the diagnostic model library
 309 should be fitted to the experimental q_e - C_e data by non-linear or weighted non-linear regres-
 310 sion (e.g. Levenberg–Marquardt or trust-region algorithms), minimising the sum of squared
 311 residuals $\text{SSE} = \sum_i (q_i^{\text{exp}} - q_i^{\text{pred}})^2$ when errors are homoscedastic. When experimental un-
 312 certainties or non-constant variance are available, weighted non-linear regression should be
 313 used with the corresponding likelihood. Linearised forms (e.g. $1/q_e$ vs. $1/C_e$ for Langmuir,
 314 $\ln q_e$ vs. $\ln C_e$ for Freundlich) must be avoided because they distort the error structure and
 315 introduce bias in the estimated parameters. Physical bounds should be imposed on all pa-
 316 rameters ($Q_{\max} > 0$, $K > 0$, $0 < m \leq 1$ for Sips, etc.), and multiple initial guesses should
 317 be tested to guard against local minima. Numerical convergence alone is not considered
 318 model acceptance. The full model library is used as a diagnostic screen; final interpretation
 319 is restricted to models that are physically admissible, identifiable from the available data,
 320 and statistically plausible.

Table 4: Model status used in the diagnostic fitting workflow.

Status	Meaning	Used for reporting?
Fitted	Numerical convergence was achieved	No
Admissible	Parameters and limits are physically meaningful	Maybe
Identifiable	Parameters are constrained by the data and bootstrap-stable	Maybe
Plausible	Within statistical support, e.g. $\Delta\text{AIC}_c \leq 2$	Yes, as candidate
Reported	Simplest physically meaningful model or descriptor range	Yes

5.1.1 Residual error metrics

No single error metric captures all aspects of fit quality; therefore, multiple complementary metrics should be reported. The complete equations are given in Section S-2.1; here we summarise their purpose and interpretation.

- **SSE** (sum of squared errors, Eq. S-106): the objective function minimised during fitting. Sensitive to large deviations (outliers).
- R^2 (coefficient of determination, Eq. S-107): fraction of variance explained. Values close to 1 indicate a good fit, but R^2 alone cannot distinguish overfitting.
- R^2_{adj} (adjusted R^2 , Eq. S-108): penalises model complexity by accounting for the number of parameters p . Preferred over R^2 when comparing models with different numbers of parameters.
- **EABS** (sum of absolute errors, Eq. S-109): less sensitive to outliers than SSE. Useful as a complementary absolute-scale measure.
- **RESID** (mean absolute residual, Eq. S-110): average deviation per data point in the same units as q_e . Provides an intuitive measure of typical prediction error.
- **ARED** (average relative error in percent, Eq. S-111): normalises each residual by q_i^{exp} , giving equal weight to low- and high-concentration points. Particularly informative for data spanning several orders of magnitude.

- **MPSED** (median percentage squared error, Eq. S-112): a robust (median-based) relative error metric, resistant to individual outliers.

- **HYBRID** (Eq. S-113): a compromise between SSE and relative error that penalises underfitting at low concentrations, where q_i^{exp} is small and squared residuals are amplified by the $1/q_i^{\text{exp}}$ weight. It also corrects for degrees of freedom ($n - p$).

We recommend reporting at least SSE, R_{adj}^2 , ARED, and HYBRID for every model. The combination provides sensitivity to both absolute and relative deviations, robustness to outliers, and a degrees-of-freedom correction.

5.2 Information criteria: AIC and BIC

In adsorption studies, model selection is often based on goodness-of-fit metrics such as the coefficient of determination (R^2) or the sum of squared errors (SSE). However, these metrics systematically favor models with a larger number of adjustable parameters, even when the improvement in fit is not physically meaningful. This is particularly problematic when comparing isotherm models of different complexity, such as Langmuir, Sips, or multi-site formulations, where additional parameters can artificially improve the apparent agreement with experimental data.

To address this issue, information criteria provide a principled framework for balancing model fidelity and complexity. These criteria penalize overparameterized models, allowing comparison across non-nested models with different numbers of fitted parameters.

The Akaike Information Criterion (AIC), its small-sample correction (AIC_c), and the

Bayesian Information Criterion (BIC) are defined as:

$$\text{AIC} = n \ln \left(\frac{\text{SSE}}{n} \right) + 2p, \quad (30)$$

$$\text{AIC}_c = \text{AIC} + \frac{2p(p+1)}{n-p-1}, \quad (31)$$

$$\text{BIC} = n \ln \left(\frac{\text{SSE}}{n} \right) + p \ln n, \quad (32)$$

where n is the number of data points and p the number of fitted parameters.

The first term in each expression reflects the goodness of fit through the residual variance, while the second term introduces a penalty that increases with model complexity. AIC applies a constant penalty per parameter, whereas BIC imposes a stronger penalty that scales with $\ln n$, favoring simpler models as the dataset grows.

For small datasets, which are common in adsorption experiments, the corrected form AIC_c is preferred, as it compensates for finite-sample bias and prevents the selection of overly flexible models. AIC_c is meaningful only when $n > p + 1$; models violating this condition are rejected as underdetermined. For weighted fits, the corresponding weighted log-likelihood should be used instead of the unweighted SSE expression.

Models are ranked according to their relative information loss, quantified by the difference

$$\Delta \text{AIC}_c = \text{AIC}_{c,i} - \text{AIC}_{c,\min}. \quad (33)$$

The conventional interpretation is that $\Delta \text{AIC}_c < 2$ indicates substantial support for multiple models, $4 < \Delta \text{AIC}_c < 7$ indicates considerably less support for the higher-AIC model, and $\Delta \text{AIC}_c > 10$ indicates essentially no support.

When several models fall within $\Delta \text{AIC}_c < 2$, model discrimination becomes ambiguous. In such cases, Akaike weights (Eq. S-132) can be used to estimate the relative likelihood of each model given the data. These weights provide a probabilistic interpretation of model support and can be used to perform model-averaged predictions, thereby reducing sensitivity

378 to model selection.

379 Importantly, the use of information criteria does not imply that one model is the “true”
 380 model, but rather identifies the model that achieves the best trade-off between accuracy
 381 and parsimony within the candidate set. In the context of adsorption isotherms, this is
 382 essential to avoid overinterpreting parameters obtained from overly flexible models that fit
 383 noise rather than underlying adsorption behavior.

384 5.3 F-test for nested models

385 When a simpler model is nested within a more complex one (e.g. Langmuir $\subset$ Sips, Langmuir
 386 $\subset$ Double Langmuir), the F-test determines whether the additional parameters provide a
 387 statistically significant improvement:

$$F = \frac{(\text{SSE}_1 - \text{SSE}_2)/(p_2 - p_1)}{\text{SSE}_2/(n - p_2)}. \quad (34)$$

388 The F-test is treated as an auxiliary criterion because nested comparisons can be delicate
 389 when the additional parameter lies near a boundary, such as the Sips limit $m = 1$. Bootstrap
 390 identifiability and physical admissibility should confirm any F-test-based decision. Beyond
 391 the F-test, physically motivated criteria can guide the choice between nested alternatives:

- 392 • **Langmuir vs. Sips:** Accept Langmuir if the Sips-derived intrinsic heterogeneity is
 393 negligible, $\sigma_H^{(\text{Sips})} = (\pi/\sqrt{3})RT\sqrt{1/m^2 - 1} \leq \alpha RT$, or if m is statistically indistin-
 394 guishable from 1 within uncertainty (Eq. S-139).
- 395 • **Langmuir vs. Double Langmuir:** Accept Double Langmuir if the two site classes
 396 are resolvable, $\Delta E_{12} = RT|\ln(K_1/K_2)| > \beta RT$ with $\beta \approx 2\text{--}3$ (Eq. S-140).
- 397 • **Sips vs. Double Langmuir:** Prefer Sips when the site-energy separation is smaller
 398 than the distribution width, $\Delta E_{12} < 2\sigma_H^{(\text{Sips})}$ (Eq. S-141).

399 The overlap coefficient (OVL) (Eq. S-136) provides a complementary, distribution-level sim-
 400 ilarity metric. A complete decision tree integrating all these criteria is given in Figure S-1.

401 5.4 Downstream descriptor use: k -Nearest Neighbours classifica- 402 tion

403 The KNN and SVM examples are intended as downstream demonstrations of how the stan-
 404 dardized descriptor table can be used once a sufficiently large labelled dataset exists. They
 405 are not used here to claim predictive performance of a trained classifier.

406 For classification and clustering, the reporting fingerprint is converted into the standard-
 407 ized feature vector

$$\mathbf{x} = (Q_{\max}, \ln K_{\text{aff}}^*, \sigma_H). \quad (35)$$

408 Here, $Q_{\max}$ captures capacity, $\ln K_{\text{aff}}^*$ captures standard-state affinity on an additive scale,
 409 and σ_H captures the reported model-implied heterogeneity.

410 Once the feature vector in Eq. (35) has been extracted for a collection of adsorbent-
 411 adsorbate systems, materials can be classified in this descriptor space without imposing a
 412 parametric model on the group boundaries. The k -Nearest Neighbours (KNN) algorithm
 413 assigns a query material to the class most common among its k closest neighbours:

$$\hat{y}(\mathbf{x}_q) = \arg \max_c \sum_{i \in \mathcal{N}_k(\mathbf{x}_q)} \mathbf{1}[y_i = c], \quad (36)$$

414 where $\mathcal{N}_k(\mathbf{x}_q)$ denotes the set of k nearest neighbours of the query point $\mathbf{x}_q$ in the standardised
 415 descriptor space, and $\mathbf{1}[\cdot]$ is the indicator function. Because KNN is non-parametric and
 416 requires no assumption about group geometry, it is well suited for exploratory classification
 417 tasks such as:

- 418 • **Performance grouping:** sorting adsorbents into high-, medium-, and low-capacity
 419 tiers based on $(Q_{\max}, \ln K_{\text{aff}}^*)$.

- **Material family identification:** given a reference library, predicting whether a new material behaves like an activated carbon, zeolite, MOF, or biochar.
- **Pollutant selectivity:** classifying adsorbent–adsorbate pairs by target contaminant class (e.g. heavy metals, dyes, pharmaceuticals, PFAS).
- **Homogeneity screening:** separating materials with near-homogeneous surfaces ($\sigma_H \approx 0$) from those with broad energy distributions ($\sigma_H \gg RT$).

Figure 4 illustrates the first of these tasks for three synthetic performance groups projected onto the $(Q_{\max}, \ln K_{\text{aff}}^*)$ plane.

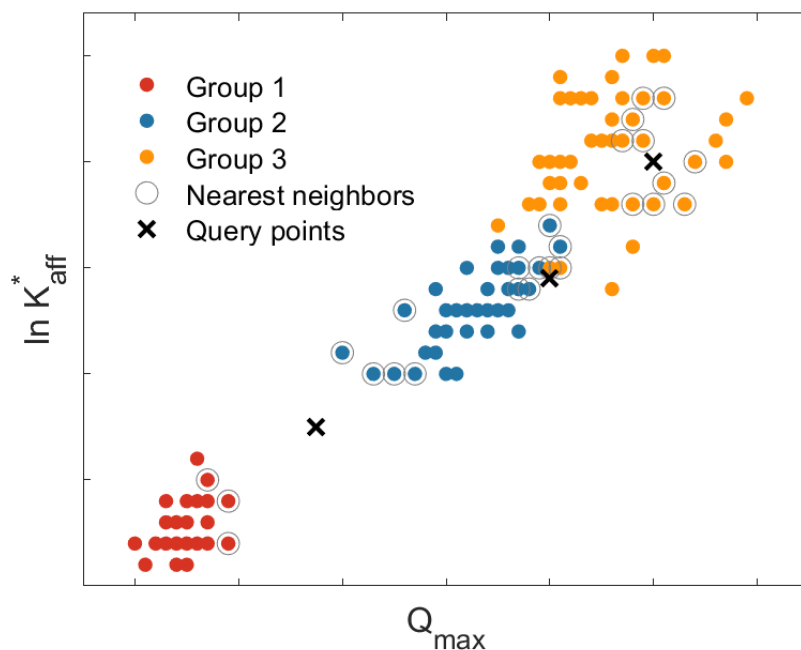


Figure 4: k -Nearest Neighbours classification of adsorbent materials in the standardized descriptor space $(Q_{\max}, \ln K_{\text{aff}}^*)$. Three performance groups are shown: Group 1 (low capacity), Group 2 (intermediate), and Group 3 (high capacity). Query points ($\times$) are classified by majority vote among their $k = 10$ nearest neighbours (grey circles). **Note:** These synthetic data are shown only to illustrate the classification workflow and should not be interpreted as classifier validation

5.5 Material classification by Support Vector Machines

When the classification task requires a well-defined decision boundary, a Support Vector Machine (SVM) finds the maximum-margin separating hyperplane in descriptor space:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C_{\text{reg}} \sum_i \xi_i, \quad y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \xi_i, \quad (37)$$

where $\mathbf{w}$ is the normal to the hyperplane, b is the bias, ξ_i are slack variables for misclassified points, and C_{reg} controls the trade-off between margin width and classification error. The support vectors—the data points that lie on or within the margin—define the boundary entirely. Figure 5a shows a linear SVM separating two classes of adsorbents in the $(Q_{\text{max}}, \ln K_{\text{aff}}^*)$ plane, corresponding to a two-dimensional projection of the feature vector in Eq. (35).

When the full harmonized descriptor set $\{Q_{\text{max}}, \ln K_{\text{aff}}^*, \sigma_H\}$ is used, the SVM operates in three dimensions and the decision boundary becomes a *plane* (Figure 5b). For non-linearly separable data an RBF kernel $\mathcal{K}(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$ replaces the linear kernel, producing curved decision surfaces that can capture complex group boundaries while still being defined entirely by the support vectors.

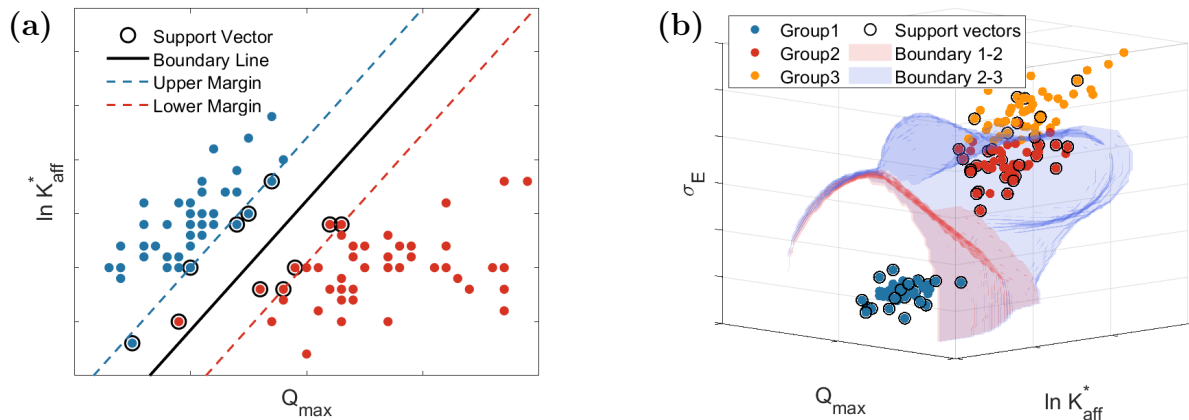


Figure 5: Support Vector Machine classification of adsorbent materials. (a) Two-dimensional projection onto the $(Q_{\max}, \ln K_{\text{aff}}^*)$ plane: the solid line is the decision boundary $\mathbf{w} \cdot \mathbf{x} + b = 0$; dashed lines indicate the margin boundaries; support vectors are circled. (b) Three-dimensional descriptor space $(Q_{\max}, \ln K_{\text{aff}}^*, \sigma_H)$: decision boundaries are now planes (or curved surfaces when an RBF kernel is used). Three performance groups and two inter-group boundaries are shown; support vectors are marked with black circles. **Note:** These synthetic data are shown only to illustrate the classification workflow and should not be interpreted as classifier validation

442 Categorical variables—such as material type (e.g. activated carbon, zeolite, MOF), best-
 443 fit isotherm model, or synthesis route—can be incorporated into the SVM framework through
 444 one-hot encoding. Each category is expanded into binary indicator features that enter $\mathbf{x}_i$
 445 alongside the continuous descriptors $(Q_{\max}, \ln K_{\text{aff}}^*, \sigma_H)$. In the resulting augmented feature
 446 space the SVM decision boundary is no longer a plane but a *hyperplane* whose dimensionality
 447 equals the total number of features minus one. For a problem with p continuous descriptors
 448 and q one-hot columns the separating surface lives in $\mathbb{R}^{p+q}$ and its projection onto any two-
 449 dimensional descriptor pair produces the familiar boundary visualised in Figure 5a. Typical
 450 SVM classification tasks in adsorption studies include:

- 451 • **Binary screening:** suitable vs. unsuitable adsorbents for a given application (e.g.
 452 materials exceeding a regulatory removal threshold for drinking water).
- 453 • **Model-type separation:** distinguishing systems that are best described by a Langmuir-
 454 type isotherm ($\sigma_H \approx 0$) from those requiring a heterogeneous model (Sips, Tóth).

455 • **Material-class boundaries:** separating material families (activated carbon vs. zeolite
456 vs. MOF) in the descriptor space, using one-hot encoded material type as additional
457 features.

458 • **Multi-class ranking:** extending the binary SVM via one-vs-one or one-vs-all strate-
459 gies to classify adsorbents into three or more performance groups (as in Figure 5b).

460 Both KNN and SVM operate on the same standardized descriptor table (Eq. S-191)
461 with Euclidean distance (Eq. S-193). Ward’s hierarchical clustering (Eq. S-199) provides
462 an unsupervised alternative when class labels are unknown. The key advantage is that all
463 three methods operate on model-harmonized descriptors, ensuring that the grouping reflects
464 standardized model-derived adsorption behavior rather than artefacts of different fitting
465 models. The full mathematical details of all clustering and classification methods are given
466 in Section S-5.

467 6 Recommended reporting guidelines

468 The following guidelines distill the preceding analysis into a practical reporting framework.
469 Their goal is not to prescribe a single isotherm model but to ensure that, regardless of which
470 model is selected, the published results are sufficient for cross-study comparison.

471 6.1 Minimum reporting set

472 For every adsorbent–adsorbate system analysed, the following quantities should be reported:

- 473 1. **Experimental conditions:** temperature T , pH, ionic strength, and the declared
474 standard-state concentration C^0 (with units).
- 475 2. **Model fit:** best-fit model name, all fitted parameters θ with units and confidence
476 intervals, and at least SSE, R^2 , ARED, and AIC_c.

3. **Reported descriptors:** $Q_{\max}$ (mg g^{-1}), K_{aff}^* (dimensionless), E_{ads} (kJ mol^{-1}), and σ_H (kJ mol^{-1}).

4. **Comparison metrics** (when applicable): OVL between competing model distributions and/or cluster labels from KNN/SVM classification.

The complete minimum reporting table is given in Table S-3; the extraction formulas for each model are collected in Table S-4.

6.2 Recommended analysis workflow

We recommend a seven-step pipeline:

1. **Data collection.** Record q_e - C_e pairs at constant T and pH, with at least $n \geq 8$ points as an absolute minimum, and preferably $n \geq p + 5$ for every candidate model considered, spanning the linear, transition, and plateau regions whenever saturation descriptors are to be reported.

2. **Non-linear fitting.** Fit all models as a diagnostic library, but interpret only the models that are physically admissible, statistically supported, and identifiable from the available data. (Section 5.1).

3. **Error metrics.** Compute SSE, R^2 , R_{adj}^2 , ARED, and HYBRID for each model (Section S-2.1).

4. **Model selection.** Rank by AIC_c ; apply F-tests for nested pairs; apply physical decision criteria (Section 5.3, Figure S-1).

5. **Descriptor extraction.** From the selected model, compute $Q_{\max}$, K_{aff}^* , E_{ads} , and σ_H (Table 2).

6. **Statistical comparison.** Compute OVL between energy distributions of alternative models or between different materials (Section S-3.2).

7. **Classification.** Map materials into the standardised descriptor space and apply clustering or supervised classification (Section 5.4, Section 5.5).

6.3 Practical recommendations

- **Never linearise.** Linearised isotherm forms distort the error structure and bias the fitted parameters. Always use non-linear regression.
- **Do not report $Q_{\max}$ from the Freundlich equation.** The Freundlich isotherm has no saturation; any “capacity” derived from it is a concentration-window artefact.
- **Always declare C^0 .** The numerical value of E_{ads} depends on C^0 ; omitting it renders the result non-reproducible. Differences ΔE_{ads} between materials measured under the same C^0 are invariant.
- **Report confidence intervals.** Bootstrap or Monte Carlo uncertainties on $Q_{\max}$, E_{ads} , and σ_H should accompany every table entry.
- **Use σ_H to assess model necessity.** If the best 3-parameter model yields only weak heterogeneity, e.g. $\sigma_H \leq 2RT$ or, for Sips, m remains close to 1 within uncertainty, the Langmuir model is likely sufficient and the extra parameter is not justified (Eq. S-139).
- **Prefer the simplest adequate model.** Among models with $\Delta\text{AIC}_c < 2$, report the one with the fewest parameters.

7 Conclusions

Adsorption isotherms remain indispensable tools for evaluating materials for water-quality applications, but their fitted parameters cannot be treated as universally interchangeable quantities. This work shows that many common comparisons in the adsorption literature mix parameters with different dimensions, different limiting behavior, and different physical

522 meaning. In particular, a maximum capacity should only be reported when the selected
 523 model contains a finite saturation plateau. Non-saturating models such as Freundlich may
 524 be useful empirical descriptions over a limited concentration range, but they do not define
 525 an intrinsic $Q_{\max}$ and should not be used to rank materials by maximum uptake.

526 To address this problem, we proposed a unified reporting framework that separates model
 527 fitting from descriptor extraction. The final result of an isotherm analysis should be reduced,
 528 whenever possible, to a common descriptor set:

$$\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}.$$

529 Here, $Q_{\max}$ describes the finite adsorption capacity, K_{aff}^* places affinity on a dimensionless
 530 standard-state scale, E_{ads} expresses this affinity as an adsorption-energy descriptor, and σ_H
 531 quantifies the reported model-implied surface heterogeneity. This fingerprint allows adsorp-
 532 tion data obtained from different isotherm models to be compared in a physically meaningful
 533 way, provided that the standard-state concentration C^0 , experimental conditions, fitted pa-
 534 rameters, and uncertainty metrics are explicitly reported.

535 The framework also provides a practical route for model selection. Rather than relying
 536 on R^2 alone, the proposed workflow combines non-linear regression, residual-based metrics,
 537 information criteria, nested-model tests, and physical validation based on limiting behavior
 538 and energy-distribution families. This approach helps distinguish cases where Langmuir is
 539 sufficient from cases where heterogeneous models such as Sips, Tóth, UNILAN, Jovanovic, or
 540 double Langmuir are genuinely justified. It also prevents over-interpretation of small statis-
 541 tical improvements that do not correspond to physically meaningful differences in adsorption
 542 behavior.

543 Finally, the descriptor-based structure of the framework makes adsorption data more
 544 suitable for cross-study comparison, uncertainty analysis, and machine-learning classifica-
 545 tion. By converting model-specific fitting results into a common set of thermodynamic and

546 heterogeneity descriptors, adsorption studies can move from isolated parameter tables to-
547 ward reproducible structure–performance relationships across adsorbents, contaminants, and
548 water matrices. The proposed guidelines are therefore not intended to impose a single pre-
549 ferred isotherm model, but to ensure that whichever model is used, the reported quantities
550 remain transparent, comparable, and physically interpretable.

551 Acknowledgement

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553 Supporting Information Available

554 The Supporting Information contains the full isotherm equations and limiting cases, residual
555 error metrics, information criteria, model-selection workflow, energy-distribution derivations,
556 descriptor-extraction tables, reporting protocol, and documentation of the companion note-
557 books.

558 References

- 559 (1) Progress on household drinking water, sanitation and hygiene 2000-2022: Spe-
560 cial focus on gender - UNICEF DATA. [https://data.unicef.org/resources/
561 jmp-report-2023/](https://data.unicef.org/resources/jmp-report-2023/).
- 562 (2) The United Nations World Water Development Report 2023: partnerships and co-
563 operation for water - UNESCO Digital Library. [https://unesdoc.unesco.org/ark:
564 /48223/pf0000384655](https://unesdoc.unesco.org/ark:/48223/pf0000384655).
- 565 (3) Mekonnen, M. M.; Hoekstra, A. Y. Four billion people facing severe water scarcity.
566 *Science Advances* **2016**, *2*, e1500323.

- (4) Vörösmarty, C. J.; McIntyre, P. B.; Gessner, M. O.; Dudgeon, D.; Prusevich, A.; Green, P.; Glidden, S.; Bunn, S. E.; Sullivan, C. A.; Liermann, C. R.; Davies, P. M. Global threats to human water security and river biodiversity. *Nature* **2010**, *467*, 555–561.
- (5) Blueprint for Acceleration: Sustainable Development Goal 6 Synthesis Report on Water and Sanitation 2023. <https://www.unwater.org/publications/sdg-6-synthesis-report-2023>.
- (6) Crini, G. Non-conventional low-cost adsorbents for dye removal: A review. *Bioresource Technology* **2006**, *97*, 1061–1085.
- (7) Fu, F.; Wang, Q. Removal of heavy metal ions from wastewaters: A review. *Journal of Environmental Management* **2011**, *92*, 407–418.
- (8) Ahmaruzzaman, M. Industrial wastes as low-cost potential adsorbents for the treatment of wastewater laden with heavy metals. *Advances in Colloid and Interface Science* **2011**, *166*, 36–59.
- (9) aus der Beek, T.; Weber, F.; Bergmann, A.; Hickmann, S.; Ebert, I.; Hein, A.; Küster, A. Pharmaceuticals in the environment—Global occurrences and perspectives. *Environmental Toxicology and Chemistry* **2016**, *35*, 823–835.
- (10) Thacharodi, A.; Hassan, S.; Hegde, T. A.; Thacharodi, D. D.; Brindhadevi, K.; Pugazhendhi, A. Water a major source of endocrine-disrupting chemicals: An overview on the occurrence, implications on human health and bioremediation strategies. *Environmental Research* **2023**, *231*, 116097.
- (11) Li, J.; Liu, H.; Paul Chen, J. Microplastics in freshwater systems: A review on occurrence, environmental effects, and methods for microplastics detection. *Water Research* **2018**, *137*, 362–374.

- (12) Rahman, M. F.; Peldszus, S.; Anderson, W. B. Behaviour and fate of perfluoroalkyl and polyfluoroalkyl substances (PFASs) in drinking water treatment: A review. *Water Research* **2014**, *50*, 318–340.
- (13) Buck, R. C.; Franklin, J.; Berger, U.; Conder, J. M.; Cousins, I. T.; de Voogt, P.; Jensen, A. A.; Kannan, K.; Mabury, S. A.; van Leeuwen, S. P. Perfluoroalkyl and polyfluoroalkyl substances in the environment: Terminology, classification, and origins. *Integrated Environmental Assessment and Management* **2011**, *7*, 513–541.
- (14) Glüge, J.; Scheringer, M.; Cousins, I. T.; DeWitt, J. C.; Goldenman, G.; Herzke, D.; Lohmann, R.; Ng, C. A.; Trier, X.; Wang, Z. An overview of the uses of per- and polyfluoroalkyl substances (PFAS). *Environmental Science: Processes & Impacts* **2020**, *22*, 2345–2373.
- (15) US EPA, O. PFAS Strategic Roadmap: EPA’s Commitments to Action 2021-2024. 2021; <https://www.epa.gov/pfas/pfas-strategic-roadmap-epas-commitments-action-2021-2024>.
- (16) Foo, K. Y.; Hameed, B. H. Insights into the modeling of adsorption isotherm systems. *Chemical Engineering Journal* **2010**, *156*, 2–10.
- (17) Hu, Q.; Lan, R.; He, L.; Liu, H.; Pei, X. A critical review of adsorption isotherm models for aqueous contaminants: Curve characteristics, site energy distribution and common controversies. *Journal of Environmental Management* **2023**, *329*, 117104.
- (18) Langmuir, I. THE ADSORPTION OF GASES ON PLANE SURFACES OF GLASS, MICA AND PLATINUM. *Journal of the American Chemical Society* **1918**, *40*, 1361–1403.
- (19) Freundlich, H. Über die Adsorption in Lösungen. *Zeitschrift für Physikalische Chemie* **1907**, *57U*, 385–470.

- 615 (20) Sips, R. On the Structure of a Catalyst Surface. *The Journal of Chemical Physics* **1948**,
616 *16*, 490–495.
- 617 (21) Latour, R. A. The langmuir isotherm: A commonly applied but mis-
618 leading approach for the analysis of protein adsorption behavior. *Journal*
619 *of Biomedical Materials Research Part A* **2015**, *103*, 949–958, _eprint:
620 <https://onlinelibrary.wiley.com/doi/pdf/10.1002/jbm.a.35235>.
- 621 (22) de Vargas Brião, G.; Hashim, M. A.; Chu, K. H. The Sips isotherm equation: Often
622 used and sometimes misused. *Separation Science and Technology* **2023**, *58*, 884–892,
623 _eprint: <https://doi.org/10.1080/01496395.2023.2167662>.

624 TOC Graphic

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